Suad Arzimanli

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Research Interests

Geoinformation systems and provenance-aware GeoETL; urban Earth observation, geovisualization, and GeoAI/forecasting for reproducible geospatial analysis (Python).

Education

MSc Mathematics and Computer Science *2025 – Present*
University of Łódź, Poland. Full-time studies in English.

BSc Mathematics and Computer Science *Oct 2021 – July 2025*
University of Łódź, Poland. Thesis: "Application using Geospatial Intelligence (GEOINT) to map and analyze geographic data" — web GIS for conflict-event analysis and forecasting (ACLED).

Research Experience

GeoLineage *2025 – Present*
Independent open-source research
Provenance-first GeoETL: STAC ingest → GeoParquet; per-run lineage JSON, run-to-run input diff, and minimum recompute. Validated on Element84 Earth Search and Copernicus Data Space with validation suite and reproducibility scripts. GitHub

Urban Protest-Count Forecasting *2025 – Present*
Collaborative research (Virginia Tech)
Urban-data integration for one-month-ahead forecasting of reported protest counts; first-author manuscript with S. Simsek and C.-T. Lu (see Manuscripts).

Urban Heat Island Analysis, Łódź *2025*
Independent EO study
Sentinel-2 NDVI and Landsat 8 LST on a 500 m grid; regression and spatial analysis; interactive Leaflet map with documented, reproducible workflow.

GEOINT Crisis Mapping *2024 – 2025*
BSc thesis
Web GIS for conflict-event analysis (Poland/ACLED); forecast panel (3-month moving average); temporal holdout evaluation of forecasting baselines (moving average, naive, Random Forest). ASP.NET Core, Google Maps API, Chart.js.

dr Łukasz Lechowski *2025 – Present*
Faculty collaboration — Institute of Urban Geography, Tourism and Geoinformation, University of Łódź
Recently started research collaboration; no joint paper yet.

Manuscripts

Arzimanli, S., Simsek, S., and Lu, C.-T. Integrating Urban Data for One-Month-Ahead Forecasting of Reported Protest Counts. *Manuscript prepared for submission to IEEE Big Data*. (University of Łódź; Virginia Tech.)

Professional Experience

Full-Stack .NET and GIS Developer *2023 – Present*
DZ CIS LLC — Washington, DC (Remote)

- Built geospatial apps (.NET MVC/ASP.NET Core, ArcGIS Pro / Experience Builder, Google Maps API) for municipal Tree Management, 311, and DOT/Highway operations; designed/built an end-to-end municipal asset-management web app (ArcGIS assets on Google Maps, Siebel/311 via Azure APIM, SQL inspections/work orders, Entra ID roles).
- Integrated Cityworks AMS, Siebel 311, and ArcGIS Feature Services over REST; automated service-request and inspection workflows.
- Designed/maintained enterprise geodatabases; spatial analysis and QA/QC with SQL and Python; technical lead for GIS/.NET delivery (SSO / Entra ID).

Skills

Programming / EO: Python, STAC, GeoParquet, Sentinel-2, Landsat, NDVI/LST, geospatial ETL and lineage, reproducible pipelines, temporal forecasting
Backend / web: C#, .NET (MVC, ASP.NET Core, Web API), EF Core, SQL Server, JavaScript, Google Maps API, Leaflet, Chart.js
GIS / ops: ArcGIS Pro, ArcMap, ArcGIS Online, Experience Builder, Cityworks AMS, enterprise geodatabases, 311/Siebel integrations

Languages

Azerbaijani — Native Turkish — Native English — Professional (C1/C2) German — Intermediate (A2–B1)